Leonardo Marciaga

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EDUCATION

BS, Applied Mathematics Expected May 2027
Illinois Institute of Technology
Minors in Artificial Intelligence and Statistics

- Cumulative GPA: 4.0/4.0
- Relevant coursework: ODEs and Dynamical Systems, Data Mining, Linear Optimization, Introduction to Stochastic Processes, Modern Methods in Discrete Applied Mathematics (graduate), Introduction to Algorithms (ongoing)

RESEARCH EXPERIENCE

Undergraduate Complexity Researcher June 2026 - August 2026
Santa Fe Institute

- Formulated mathematical models of group formation among heterogeneous agents from individual and collective perspectives using techniques from multi-armed bandits and evolutionary game theory, respectively.
- Programmed simulations of individual agent decision-making over 30,000 steps and 100 trials to validate theoretical results from multi-armed bandit theory.
- Analyzed the equilibrium properties and critical thresholds of the evolutionary dynamics of cooperative strategies among two subpopulations via a system of ordinary differential equations and game theory.

Research Fellow January 2026 - May 2026
SoReMo Initiative, Illinois Tech

- Developed a decision support tool using Python and Google OR-Tools library based on a constraint programming model that maximizes total assigned preference and minimizes makespan, achieving higher total assigned preference than a makespan-first baseline (8.76% on average) with no increase in tardiness.
- Identified important training components via a training needs assessment procedure based on task and knowledge, skills, and abilities (KSA) analysis.

Research Intern June - August 2025
Institute for Pure and Applied Mathematics (IPAM), UCLA

- Collaborated with the RAND Corporation in development of methodology leveraging Deep & Cross neural network architectures, LLM embeddings, and cluster-informed data generation to address challenges of survey evolution and data sparsity in the social sciences.
- Developed and validated LLM-based synthetic data generation pipeline against data quality metrics, achieving 80% overall similarity to real-world statistical trends.
- Applied clustering algorithms such as K-medoids, hierarchical, and spectral clustering, in complex longitudinal survey data encompassing nearly 10 years and 2000+ respondents to characterize positive/negative attitudes towards vaccination.
- Published at the 2026 AAAI Student Abstract and Poster Program as listed below.

Research Assistant August 2024 - Present
Department of Applied Mathematics, Illinois Tech

- Ongoing project with Professors Hemanshu Kaul (Illinois Tech) and Jeffrey Mudrock (University of South Alabama) on graph coloring applying recent methods from probability, algebra, and matching theory.
- Resulted in the solution of an open question in graph coloring using methods from graph criticality, enumerative functions and real analysis including Karamata's inequality. Resulted in a published at an international peer-reviewed journal as listed below.

PUBLICATIONS

Rezvani, J., Hyk, A., Pham, T., Marciaga, L., Liao, C., Vardavas, R., & Mitsopoulos, K. (2026). Semantic Embedding and Synthetic Augmentation for Longitudinal Survey Prediction (Student Abstract). Proceedings of the AAAI Conference on Artificial Intelligence, 40(48), 41365-41367. `https://doi.org/10.1609/aaai.v40i48.42271`

Kaul, H., Marciaga, L., and Mudrock, J. *List Coloring the Cartesian Product of a Complete Graph and Complete Bipartite Graph.* Enumerative Combinatorics and Applications, 6 (2026), article S2R5. `https://doi.org/10.54550/ECA2026V6S1R5`.

TEACHING

Associate Instructor January 2021 - Present

Panamanian Mathematical Olympiad Foundation, Panama

- Led training sessions for Panama's math contest training program, leading to award-winning students at contests such as the International Mathematical Olympiad, the Ibero-American Mathematical Olympiad, and the Pan-American Girls' Mathematical Olympiad.
- Mentored students, organized training schedules and created class material in mathematical olympiad topics, such as algebra, combinatorics, geometry, and number theory.

Teaching Assistant August - December 2024

Department of Computer Science, Illinois Tech

CS 105: Introduction to Computer Programming

- Taught basic programming concepts to 15 non-CS students, providing personalized assistance and feedback, both in-class and through homework.
- Identified areas for improvement in class performance via discussions with lead instructor and other teaching assistants.

LEADERSHIP

President of SIAM IIT Student Chapter January 2026 - Present

Illinois Tech

- Organized events for the applied mathematics community at IIT within semester-long budget periods.
- Led a 6-person executive board, delegating communications, marketing, and budget administration.
- Successfully executed 1-2 weekly events, including seminars with external speakers for audiences of approximately 10 undergraduates and social events with 50+ attendees.

HONORS AND AWARDS

Illinois Tech College of Computing Undergraduate Research Excellence Award	2026
AAAI-26 Student Scholarship & Volunteer Award	2025
Illinois Tech Clinton E. Stryker Service and Leadership Award	2025
85th William Lowell Putnam Mathematical Competition - Top 500	2024
International Mathematical Olympiad (IMO) - Honorable mention	2018 - 2020

PROFESSIONAL MEMBERSHIPS

Association for the Advancement of Artificial Intelligence (AAAI)	2025 - Present
Society for Hispanic Professional Engineers (SHPE)	2025 - Present
Society for Industrial and Applied Mathematics (SIAM)	2024 - Present

SKILLS

Programming Languages: Python, R, HTML, CSS, MATLAB, SQL, Mathematica
Machine Learning Libraries: scikit-learn, TensorFlow, PyTorch
Data Analysis & Visualization Libraries: pandas, matplotlib, tidyverse, ggplot2, plotly